3-D crystals: carbon allotropes

Assignment #4

General considerations:

- Use GGA pseudo-potentials.
- Perform spin un-polarized calculations.
- Use a Γ-centered k-mesh.
- DOS calculations: use the tetrahedron method (ISMEAR=-5 in the `INCAR` file).

Carbon allotropes: face-centered cubic, diamond and graphite

Calculate and discuss the structural and electronic ground state properties of the face-centered cubic (FCC), diamond and graphite carbon allotropes. Follow the indications in the next sections.

Carbon FCC

Construct a carbon FCC cell with volume $V = 7.44$ Å^3, and perform a convergence test on the k-point mesh.

Determine the ground state structure by calculating the total energy of the system as a function of the volume.

Calculate the electronic density of states (DOS). Analyze the electronic properties of the material, considering in particular the Fermi level.

- Use the tetrahedron method (in the `INCAR` file set ISMEAR=-5).
- Convergence test calculations on the k-point mesh: vary the number of k points along all reciprocal lattice vectors, simultaneously, starting from a $5 \times 5 \times 5$ grid; identify the smallest k mesh which satisfies the convergence criteria on the total energy of 1 meV/atom per variation of 1 k point in evry direction; identify also the amount of irreducible k points.

Carbon diamond

Construct a carbon diamond cell (by displacing two FCC lattices and by using the experimental value for the lattice vectors $a = 3.57$ Å)[1] and perform a convergence test on the k-point mesh.

[1] Landolt-Börnstein, Group III Condensed Matter, volume 23A, doi:10.1007/10377019_7

(Optional): Perform a convergence test on the Gaussian smearing, by considering 0.1, 0.05 and 0.01 eV as values for SIGMA; check the extrapolated energy (*energy(sigma->0)* in the `OUTCAR`).

Determine the ground state structure by calculating the total energy of the system as a function of the volume. Compare the results with the experimental value.

Calculate the electronic density of states (DOS). Check the robustness of the results for the DOS versus the sampling of the reciprocal space; increase the number of k points if necessary. Analyze the electronic properties of the material, considering in particular the Fermi level. Compare the DFT result with the experimental value for the band gap $E_{GAP} = 5.5$ eV.

(Optional): Calculate the DOS by using the Gaussian method, and compare the results with the data obtained by the tetrahedron smearing.

« Use the Gaussian smearing (in the `INCAR` file set ISMEAR=0 and SIGMA=0.01), except for the DOS calculation.

« Convergence test calculations on the k-point mesh: vary the number of k points along all reciprocal lattice vectors, simultaneously, starting from a $5 \times 5 \times 5$ grid; identify the smallest k mesh which satisfies the convergence criteria on the total energy of 1 meV/atom per variation of 1 k point in every direction; identify also the amount of irreducible k points.

Carbon graphite

Construct a carbon graphite cell with 4 atoms, and perform a convergence test on the k-point mesh.

Determine the ground state structure by calculating the total energy of the system as a function of the volume. Keep the interlayer distance fixed to the experimental value ($d = 3.305$ Å). Compare the results with the experimental values for the lattice vectors ($a = b = 2.439$ Å and $c = 6.610$ Å)[2]. Compare the values of the in-plane lattice vectors a and b obtained for the graphite with those obtained relaxing the graphene in the Assignment 4.

Calculate the electronic density of states (DOS). Check the robustness of the results for the DOS versus the sampling of the reciprocal space; increase the number of k points if necessary. Analyze the electronic properties of the material, considering in particular the Fermi level.

« Use the tetrahedron method (in the `INCAR` file set ISMEAR=-5).

« Convergence test calculations on the k-point mesh: setup the calculations by considering the experience acquired while studying the 2-D hexagonal lattices in the Assignment 4.

Diamond band gap

Calculate the electronic energy bands of diamond. Discuss the band gap: is the band gap direct or indirect?

[2] Hu Q., *et al.*, PRB 73, 214116 (2006)

« Use the Gaussian smearing (in the `INCAR` file set ISMEAR=0 and SIGMA=0.01).

« Perform a non-self-consistent calculation on a k path including the relevant high symmetry lines in the `KPOINTS` file [a]: set ICHARG=11 to read previously converged CHGCAR and WAVECAR files; avoid to write wave-function and charge density files (set LWAVE and LCHARG to false).

[a] W. Setyawan, S. Curtarolo, Elsevier Comp. Mat. Sc. 49, 2, 299-312 (2010)